

Pratiush Karki

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SUMMARY

Software engineer building production web applications across React, Next.js, FastAPI, Python, TypeScript, and PostgreSQL, with experience in backend systems, cloud deployment, and AI applications including LLMs and RAG. Quick learner and strong communicator with a high-ownership mindset, driven to tackle unfamiliar problems and turn ideas into reliable products.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks: React, Next.js, FastAPI, Node.js, Express, Streamlit

AI/ML: LangChain, RAG, ChromaDB, LLaMA, Ollama, scikit-learn

Data: PostgreSQL, Redis, Firebase, pandas, NumPy

Tools/Cloud: Git, GitHub Actions, VS Code, Docker, AWS EC2, Figma

EXPERIENCE

Greener Montclair · Software Engineer (Contract) · Montclair, NJ

Sep 2026 - Present

- Building a headless apparel storefront on Shopify Hydrogen (React, TypeScript) against the Storefront GraphQL API, deployed on Oxygen, with products and orders managed in the Shopify admin.

New York Jets · Data Analyst (Contract) · Florham Park, NJ

Jan 2026 - May 2026

- Led a 5-person analytics team that identified an underserved college-aged fan segment representing 62% of surveyed respondents by analyzing historical Jets data and survey responses from ~1,000 target consumers.
- Collaborated with the marketing team and came up with three growth strategies pitched to Jets executives, sizing up to ~\$110K in game-day spend per 1,000 students converted.

MicroJournal · Product Engineer Externship · Caldwell, NJ

Jan 2026 - May 2026

- Built a 0-to-1 product, MicroJournal, in a startup studio run by the CTO and CPO of Instructure, leveraging AI tools to accelerate development and testing while adhering to best SWE practices and feedback from the Instructure engineering team.
- Validated demand with a public waitlist and submitted the final product to the App Store (iOS).

Whisk It All · Freelance Web Developer · Glen Ridge, NJ

May 2025 - Aug 2025

- Rebuilt the café's web platform with a headless CMS, a GitHub-connected deployment pipeline, and CDN delivery, cutting external vendor dependency on SpotHopper by ~80%.

Caldwell University · Systems Assistant & Student Tutor · Caldwell, NJ

Feb 2023 - Feb 2025

- Tutored course material and clarified complex concepts across multiple courses: Data Structures, Algorithms, Front-End, and Full-Stack Web Development, impacting up to 1,500 students.
- Supervised audio-visual systems for major campus events with 100+ attendees, and troubleshooted AV failures in real time under live-audience pressure, keeping output uninterrupted.

PROJECTS

Lucid Tone: Real-Time Background Music Generation Platform | Python, FastAPI, WebSockets, Redis, React, WASM

- Developed a Markov-chain music engine and a rendering pipeline capable of generating audio ~100x faster than real time for smooth generative music playback.
- Managed persistent multi-session WebSocket connections with Redis-backed shared state, buffered audio delivery, and Opus encoding to reduce network bandwidth and egress.

Fully Local RAG Agent | Docker, Python, ChromaDB, Ollama, Streamlit, AWS EC2

- Built a fully local AI agent using retrieval-augmented generation (RAG) over an unstructured document corpus, returning grounded answers traceable to their source passages with no data leaving the machine.
- Containerized with Docker and deployed to AWS EC2. Built during a hackathon with a three-person team on a Git workflow with code review.

NILINK: Platform Connecting Brands and Athletes for NIL | Next.js, TypeScript, PostgreSQL, Tailwind

- Shipped a two-sided marketplace covering the full name-image-likeness deal path: campaign discovery, applications, offers, contracts, and execution. Demoed and judged at the 2026 Capstone Showcase.
- Modeled the deal lifecycle in PostgreSQL with row-level security so each party's commercial terms stay private from the other, then held the same boundary in application code through role-aware routing and protected API routes.

EDUCATION

3.85 GPA · B.S. Computer Science · Business Analytics Minor · Honors Student

Caldwell University

Research: AI Sycophancy and LLM Reliability · 2026

jstor.org/stable/community.42398302

- Investigated sycophantic behavior in RLHF-tuned language models, proposing a five-stage feedback model and mitigation strategies spanning model training, product guardrails, and interface design.

Activities & Awards: Co-founder & Secretary, Computer Science Club, University Choir Singer, Dean's List (8x), Recognition Award